

Magnetic separation

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The magnetic separation method is used to separate materials of different magnetic properties. For example, titanium ore deposits contain minerals such as zircon (ZrSiO_4), monazite ($[\text{Ce,La,Th}] \text{PO}_4$), ilmenite (FeTiO_3), rutile (TiO_2), among which the economic importance of titanium-containing minerals is difficult to overestimate. Rutile and ilmenite are used in the production of non-toxic pigments used in inks, plastics, ceramics and more. Therefore, rutile and ilmenite must be separated from titanium ores. Magnetic separation uses magnetic forces to separate magnetic materials. The operating principle of the magnetic separator is shown in Fig. 1.

Fig. 1: (1) box; (2) non-magnetic roller L_1 ; (3) magnetic roller L_2 ; (4) conveyor belt; (5) magnetic particles; (6) collection trays.

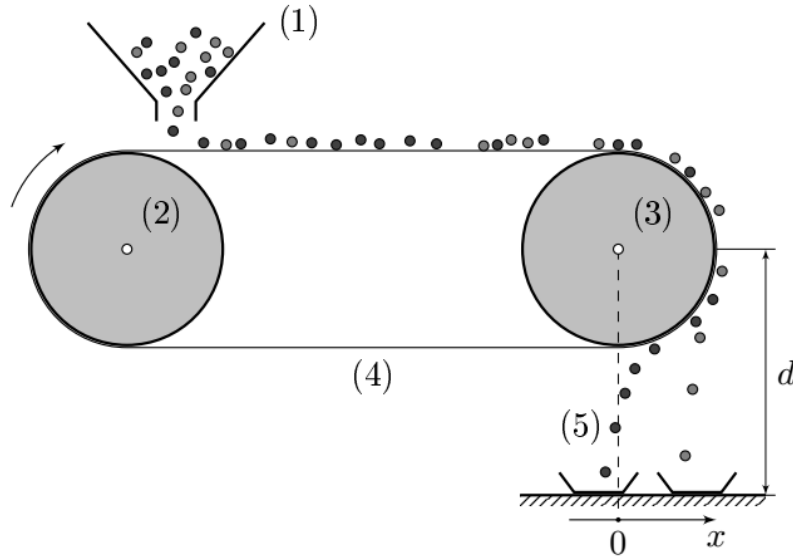


Figure 1: Figure 1.

The main part of the magnetic separator consists of two rollers L_1 and L_2 (diameter D), connected by a conveyor belt (Fig. 1). The thickness of the tape compared to the radius of the rollers can be neglected. Roller L_1 is non-magnetic. The roller L_2 is magnetic, and the magnetic field around it changes according to the following law

$$B(r) = |\vec{B}| = B_0 e^{-r/a},$$

where r is the distance from the surface of the roller in the radial direction, and a is some constant. The magnetic roller rotates with angular velocity ω , which is maintained by the electric motor. Mineral particles flow from the box onto a conveyor belt, and along it to the magnetic roller. The friction between the belt and the magnetic roller is high enough to prevent slipping between them.

The following mathematical formulas may be useful:

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \begin{cases} 1 + e^{-2x} & \text{for } x \rightarrow \infty \\ \frac{1}{x} + \frac{x}{3} - \frac{x^3}{45} & \text{for } x \rightarrow 0 \end{cases},$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \approx 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

In this part we will discuss the model of this magnetic separator, taking into account only magnetic and gravitational forces.

Part A. Magnetic susceptibility of materials

Let's study the properties of paramagnetic materials. Let us consider a mineral consisting of magnetic atoms X with concentration n , each atom has a magnetic moment p . The concentration is low, so the interaction of the magnetic moments of atoms with each other can be neglected. In the absence of an external magnetic field, the magnetic moments of the atoms are directed randomly, resulting in the magnetization $\vec{M}$ being zero. The magnetization of a substance is the magnetic moment of a unit volume of a substance. In an external magnetic field, atoms are ordered along the direction of the external magnetic field $\vec{B}$, a magnetic moment appears in the material, and non-zero magnetization is obtained. According to quantum mechanics, the projection of the magnetic moment of an atom onto the field direction (we choose the z axis along it) is equal to:

$$p_z = g_j \mu_B m,$$

where m is a number that takes values in the set $\{j, j-1, j-2, \dots, -j\}$, j is a non-negative integer or half-integer characterizing the atom X , $\mu_B = 0.93 \cdot 10^{-23} \text{ A} \cdot \text{m}^2$ is the Bohr magneton, and g_j is the so-called Landé factor. The probability $W(p_z)$ that the projection of the magnetic moment on the field direction takes the value p_z is given by

$$W(p_z) = A e^{-\frac{U_B}{kT}},$$

where A is a normalization constant, U_B is the energy of the atom in the field, T is the temperature, and k is Boltzmann's constant. Note that, by symmetry, the average projections of the magnetic moment on the x and y axes are equal to zero.

A1 Find the average value $\langle p_z \rangle$ of the component of the magnetic moment of an atom X on the axis z located in the magnetic field $\vec{B}$ at temperature T , as a function of the parameter $\alpha = \frac{g_j \mu_B B}{kT}$. **1.10**

A2 Give an expression for $\langle p_z \rangle$ in the limiting case $\alpha \rightarrow 0$ (that is, for cases where the magnetic energy is much less than the thermal energy). Consider also the case $\alpha \rightarrow \infty$. Provide a schematic graph of the dependence of $\langle p_z \rangle$ on the parameter α . **1.10**

Magnetic susceptibility χ characterizes the magnetic properties of a substance. In a weak magnetic field $\vec{B}$ the susceptibility is defined as $\chi = \frac{\mu_0 M}{B}$, where M magnetization, μ_0 is the magnetic permeability of the vacuum. Consider the magnetic field weak, that is, the parameter $\alpha \ll 1$.

A3 Calculate the magnetic susceptibility χ . Express your answer in terms of atomic concentration n , temperature T , Lande factor g_j , parameter j , and fundamental constants. **0.90**

Part B. Magnetic separator

Let's look at the model of magnetic separators described above. The dependence of the magnetic field induction B at a point on the distance r from this point to the surface of the magnetic roller along the radial direction is presented in the table:

r , mm	B , T	r , mm	B , T	r , mm	B , T
1	0.606	5	0.098	9	0.024
2	0.415	6	0.070	10	0.014
3	0.262	7	0.045	11	0.008
4	0.144	8	0.030	12	0

B1 Using the data in the table, determine the numerical values of the parameters B_0 and a from equation (1). Write down the formulas by which you determined them. **0.80**

Hereinafter, for calculations, use the following numerical values of the problem parameters:

- $D = 0.04$ m
- $R_0 = 0.5 \cdot 10^{-3}$ m
- $a = 3.0 \cdot 10^{-3}$ m
- $B_0 = 1.0$ T
- $d = 1.0$ m
- $\omega_0 = 2\pi \cdot 240$ min⁻¹
- $\rho_1 = 5.0 \cdot 10^3$ kg/m³
- $\rho_2 = 4.3 \cdot 10^3$ kg/m³
- $\chi_1 = 6.3 \cdot 10^{-4}$
- $\chi_2 = 6.5 \cdot 10^{-5}$
- $\Delta = 0.2$ m

Let us consider particles from one paramagnetic mineral with an average magnetic susceptibility χ_1 and density ρ_1 . Consider the particles to be spherical. Also consider that the magnetic moments of particles placed in the magnetic field of the roller are located at their center. The particle radius is R_0 . The collection tray is located under the magnetic roller at a distance d from it (Fig. 1). The angular velocity of the roller is ω_0 . The interaction between the magnetic moments of particles can be neglected. The magnetic force acting on the particles that left the roller can also be neglected. When the magnetic field induction changes by a small amount dB the change in particle energy is written as $dE = -\vec{P} \cdot d\vec{B}$, where $\vec{P}$ is the magnetic moment of the particle.

B2 What is the energy of the particle in the field $\vec{B}$? Express the answer in terms of the magnetic field induction B , the magnetic susceptibility of the particle material χ_1 and its volume V . **0.80**

B3 Find an expression for the position of the tray x in which it collects mineral particles. Find the numerical value for x . **1.10**

Let us assume that the mineral particles are spheres of different radii. The relative deviation of the radius from the average does not exceed 10%, that is, $\left| \frac{R_0 - R}{R_0} \right| \leq 0.1$, where R and R_0 are the radius of a particular particle and the average radius of all particles, respectively. Angular speed of the roller ω_0 .

B4 Find the expression and numerical value for the width of the particle beam Δ_0 in the direction of the axis x near the tray. **1.10**

A mixture of particles consisting of two different minerals with magnetic permeabilities χ_1 and χ_2 and densities ρ_1 and ρ_2 , respectively. Consider the average particle radii of two different minerals to be R_0 .

B5 Find an expression for the angular velocity of the magnetic roller ω , at which particles of two types can be obtained separately. Find the numerical value of ω . Assume that the width of both trays Δ is known. **1.10**

Source: <https://pho.rs/p/3049>